Asset prices and Growth

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Outline

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2. The determination of the long run real rate
   - Preliminaries: Matching theory to data
   - Theory review (Baker et al)
   - Beyond the standard framework
3. Empirical considerations
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Motivation

- Debate: Why has the real rate of interest on "safe" assets been declining in recent decades?
- Considerable practical as well as policy interest. For instance:
  - Future path of interest rates of interest if you happen to be an investor.
  - Concern until recently (and maybe again in the near future?): Zero nominal interest often viewed as a "boundary"; If $i < 0$ there is some uncertainty whether further reductions will in fact stimulate the economy (e.g., "savings paradox")
  - Low interest rates makes debt "cheap", which might influence firm dynamics (and government) behavior.

Today we will think about asset prices in the sense of <u>the real rate of interest</u>.

Preliminaries

- If we are interested in understanding the determination of the long-run real rate of interest, it needs to be endogenous $\Rightarrow$closed economy perspective.
- Which closed economy? (and what interest rate, btw?).
  - Large countries? Probably not in the current regime (globalized movements of capital). But would possibly be relevant during the years between, say, WWI and WWII.
  - In the current world: The entire world then? (absent trade with Mars, that should work). Drawback: Very high level of aggregation.
  - Key thing is to identify regions that over longer period of time is roughly in current account balance or, where $S = I$. Turns out that something slightly less extreme than the entire planet" will do, if we focus on the last roughly 40 yrs or so..

Preliminaries

Figure 5: Current account in advanced economy block and in selected individual economies

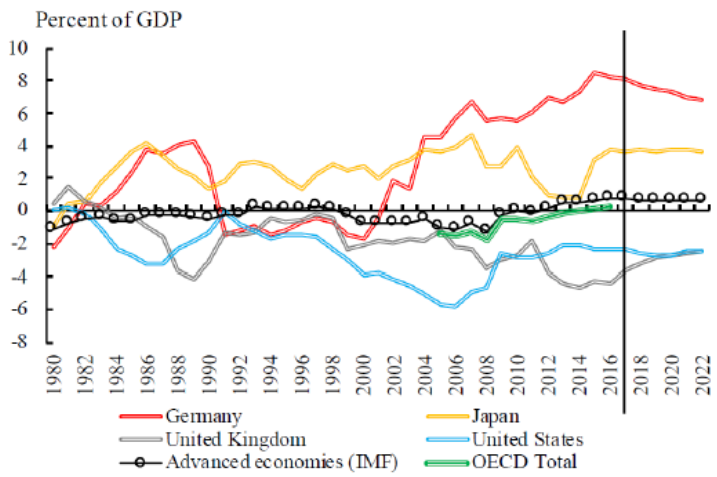


Figure: Rachel and Summers (2019)

- The block of (39) countries labelled "advanced economy block" by IMF fits the bill. $S = I$ not a bad first approximation (*why* this has been the case is a bit of a puzzle, probably related to lit going back to Feldstein-Horioka puzzle).

Preliminaries

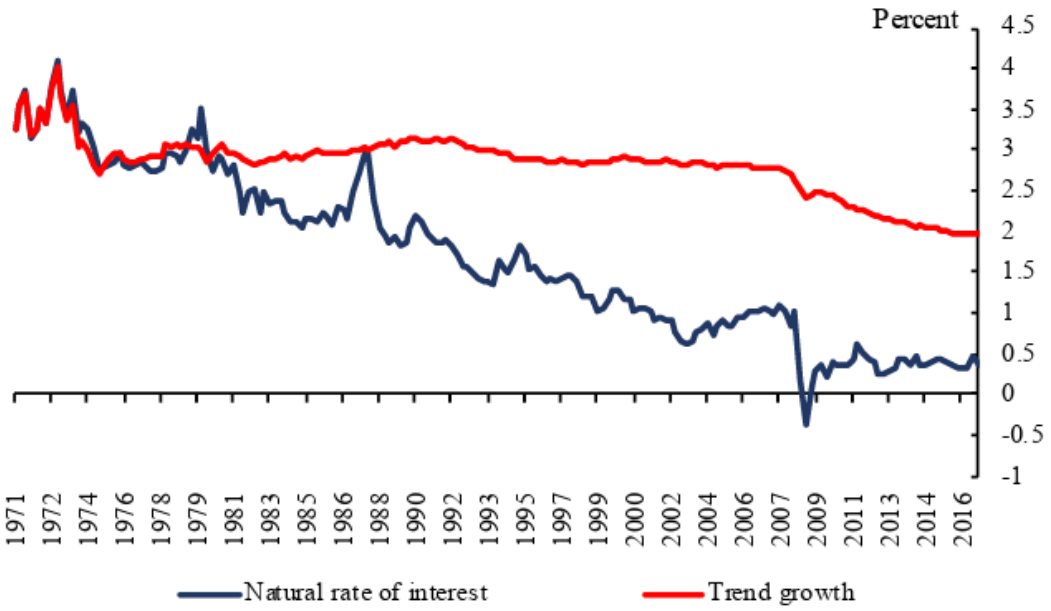
- What real rate ? The structural real rate of interest that central banks have in mind when they invoke the Taylor principle

$$i = \underbrace{r^* + \pi}_{i^*} + a(\pi - \pi^*) + b(\ln y - \ln y^*)$$

  The real rate is sometimes subsumed as the constant term in the equation to be estimated along with a and b; π^* is the target inflation rate (ECB = 2 percent).
  - Labelled natural real rate, or simply r^* in most applied work. That is, the real rate which is consistent with stable inflation and output at its potential level (cf Talyor equation).
- A meaningful theoretical counterpart to this empirical object is the steady state real rate of interest in growth theory: No inflation (money neutrality assumed) and no bussiness cycle fluctuations (thus, output at potential). This is also how central banks think about it, if you pick up a report.

Preliminaries

- Since at least the 1970s the natural real rate has been declining, which is viewed as the underlying reason why other rates (prominently the government bond rate) has been declining



Source: Rachel and Summers (reading list) employing estimation approach of Laubach and Williams (2003).

Theory review

- We are equipped with a series of standard models, which hold predictions for the evolution of real rate of interest.
- Most fundamental is the *Solow-Swan model*. Consider a version where the representative firm uses a Cobb-Douglas production function:

$$Y = K^{\alpha}(AL)^{1-\alpha} \Longleftrightarrow Y = AL\left(\frac{K}{Y}\right)^{\frac{\alpha}{1-\alpha}} \equiv AL\kappa^{\frac{\alpha}{1-\alpha}}$$

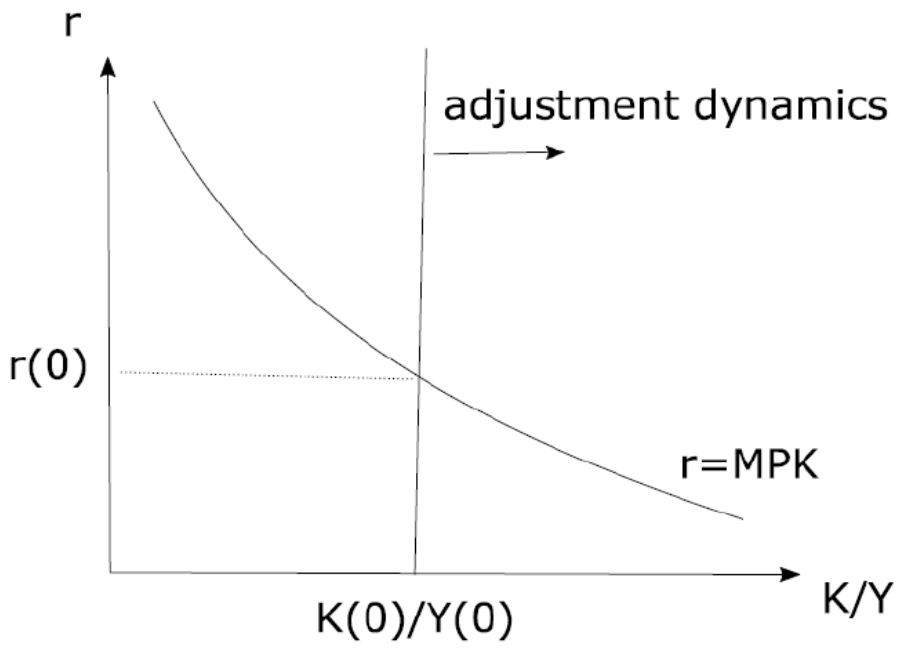
- Law of motion in the capital-output ratio (I am ignoring depreciation, $\delta = 0$)

$$\frac{\dot{\kappa}}{\kappa} = \frac{\dot{K}}{K} - \frac{\dot{Y}}{Y} = s\frac{Y}{K} - \frac{\dot{Y}}{Y} = s\kappa^{-1} - \frac{\alpha}{1-\alpha}\frac{\dot{\kappa}}{\kappa} - g - n \qquad (1)$$

- or

$$\dot{\kappa} = (1-\alpha)\,s - (1-\alpha)\,(n+g)\,\kappa \equiv \phi(\kappa)$$

Theory review



- Beneath the surface. The instantanous determination of the real rate, and dynamics towards st.st.

Theory review

- To make this more explicit. From the firm problem we know (depreciation is ignored)

$$r = \alpha\left(Y/K\right) = \alpha\kappa^{-1}$$

- From which follows $\dot{r}/r = -\dot{\kappa}/\kappa$, and thus (substituting for κ and $\dot{\kappa}$ in the law of motion above):

$$\frac{\dot{r}}{r} = (1-\alpha)\left(n+g\right) - \frac{(1-\alpha)}{\alpha} sr \equiv f(\kappa)$$

, which then gives the Solow-Swan models prediction regarding the path of the real rate. In the long run steady state:

$$\lim_{t\to\infty} r = r^* = \alpha\frac{n+g}{s}.$$

Theory review

- We have the standard comparative statics

$$\frac{\partial r^*}{\partial s} < 0, \frac{\partial r^*}{\partial g} > 0; \frac{\partial r^*}{\partial n} > 0; \frac{\partial r^*}{\partial \alpha} > 0$$

- Savings propensity; demography; productivity growth; production technology are thus key determinants.
- What about the endogenous growth stuff?
  - Still there embedded in "g" in the aggregate (all advanced countries, or the world); it is not exogenous. (In a small country like, say, Denmark it is probably a better approximation to view it as exogenous)
  - Policies, for instance, will then also matter to g and implicitly, to r . Determinants of policies will matter to g and thus r. (When we get to Idea - based growth determinants of r appears from here as well)
  - We return to the nature of the influence from policy below. So far, no government, and thus no policies.

Theory review

- Note, that long-term movements can also be unleashed by shocks!
  - For instance, a positive shock to the level of productivity will obviously increase the marginal product of capital in the short run. This will increase r^* , which then subsequently falls back to its orginal level (capital accumulation does the adjustment). Unleashes what would appear to be a long period of decline (if you "forget" the initial shock)
  - Later in the course (Asgers part), you will see how misallocation matters to TFP. Changes in the degree of misallocation (positive or negative), potentially instigated by policy, can be represented (crudely) as a change in "A".
- The length of time outside the steady state (following a shock) depends on the rate of convergence. Barros classic "iron law of convergence" of two percent (which is not really a law, since higher estimates exist), would imply that half the gap to steady state is closed in about 30 yrs...

Theory review

- What other theories do we have, beyond Solow-Swan? *Diamond model* (2 period OLG).
- The predictions are fairly similar to the Solow-Swan model in some dimensions (cf Baker et al):

$$k_{t+1} = \frac{\overbrace{(1-\alpha)\,k_t^{\alpha}}^{w_t}}{(2+\rho)\,(1+g)\,(1+n)} \equiv \varphi(k_t)$$

$$\Longrightarrow \left(\frac{y}{k}\right)^{*} = \frac{(2+\rho)\,(1+g)\,(1+n)}{(1-\alpha)} \Rightarrow r^{*} = \frac{\alpha\,(2+\rho)}{(1-\alpha)}\,(1+g_Y^{*})$$

Comparative statics similar to Solow-Swan (plus preferences)

$$\frac{\partial r^{*}}{\partial g} > 0;\ \frac{\partial r^{*}}{\partial n} > 0;\ \frac{\partial r^{*}}{\partial \alpha} > 0,\ \frac{\partial r^{*}}{\partial \rho} > 0$$

- These results draw on some pretty strong assumptions: Cobb-Douglas technology; log utility (u (c) = ln (c)).

Theory review

- In general the dynamics of the Diamond model can be highly complex, featuring multiple steady states and even indeterminancy (multiple capital stocks and thus interest rates that fulfill the law of motion for capital).
- For instance (utility per period is here $\frac{c^{1-\theta}-1}{1-\theta}$)and output is $y = f(k)$)

$$k_{t+1} = \frac{1}{(1+g)(1+n)} \frac{\overbrace{f(k_t) - k_t f'(k_t)}^{w_t}}{1 + (1+\rho)^{1/\theta} \underbrace{(1 + f'(k_{t+1})}_{(1+r_{t+1})^{\frac{\theta-1}{\theta}}})^{\frac{\theta-1}{\theta}}} \equiv \varphi(k_t, k_{t+1})$$

- k_{t+1} appears on both sides! Expected real rate during old age matters for savings today, which determines the real rate in old age. Not explored in applied work (to my knowledge)

Theory review

- A third standard model is the *Ramsey-Cass-Koopmans model*. The simplest way to elicit predictions is via the Consumption Euler
$$\frac{\dot{c}}{c} = \frac{1}{\theta}(r - \rho)$$
- In the steady state per capita consumption grows at the rate g (exogenous technological change). Putting $\frac{\dot{c}}{c} = g$ and rearranging terms
$$r^* = \theta g + \rho$$
- Similar to Diamond in terms of preferences and g.
- No demography? This actually comes from assumptions about utility. Requires $U_0 = \int u(c) L e^{-\rho t} dt$. Mechanism yielding zero impact of n on r? No impact from factor income distribution (i.e., α).

Theory review:Government

- RCK: No role of debt (Ricardian equivalence), nor G if financed by undistorting (say, lump-sum) taxes in the steady state, See Barro & Sala-i-Martin, ch. 3.
- Diamond: Crowding-out, even with lump sum taxes (no ricardian equivalencet).
- RCK: Distorting taxes? Capital taxes raises r^* . Labor taxes has no effect on accumulation.
- Diamond: Capital taxes will not influence r^* as long a savingsrate is insensitive to real rate (log utility, for example). Labor taxes will increase r^*
  - These differences are ultimately due to differences in the savings motive (altruistic bequest vs life cycle)
- Note: in effect distributional impact on young vs old in OLG economy: wage tax affect the young (saving type), capital tax the old (dissaving). Government policy that redistribute between agents with different MPCs (old have a MPC=1).

Beyond the standard framework

- Inequality Rich save more. Rising inequality raises savings, and brings down r
- Savings glut hypothesis. Influence from emerging economies. Probably not key, cf. "S=I figure" from Rachel and Summers.
- The relative price of capital goods. Ambigious: Less savings are needed for a given amount of extra equipment (lower investment demand, lower r); Cost of investment declines inticing to more capital investments (more investment demand, higher r). If easy to substitute capital and labor the latter dominates. Empirically, lower elasticities seem to prevail ($E < 1$), but controversial issue.
- Gov'ment affects extent precautionary savings (if influence on percieved risk at household level).
- Declining public investments?
- Money non-neutrality? (See Reis, 2007, Jorda et al., 2020)

Beyond the standard framework

In Dalgaard and Olsen (forthcoming) we note that capital depreciation has been on the rise (here: US)

- A decomposition analysis show that it is *within industry movements* that drives it. Specifically, changes in the composition of the capital stock towards more short lived assets

Beyond the standard framework

- "Depreciation" can be understood as either physical (δ in standard growth models) or economic. Cannot be seperated in the data.
  - Since ICT capital drives a lot of the observed increase economic depreciation probably more important that physical in recent years.
- Seperation not nessesary to gauge the influence. FOC from firm optimization where p is the price of investment (output price normalized to 1).

$$F_K = p_{t-1}\left[1 + r_t - \frac{p_t}{p_{t-1}}(1-\delta)\right].$$

- Key observation: Higher δ (physical depreciation) **or** faster investment price *deflation* ($p_t < p_{t+1}$; economic depreciation) *increases* the (user-)cost of capital.
  - The above is two sector setting. In the one sector setting the FOC collapes to: $F_K = MPK = r_t + \delta$.

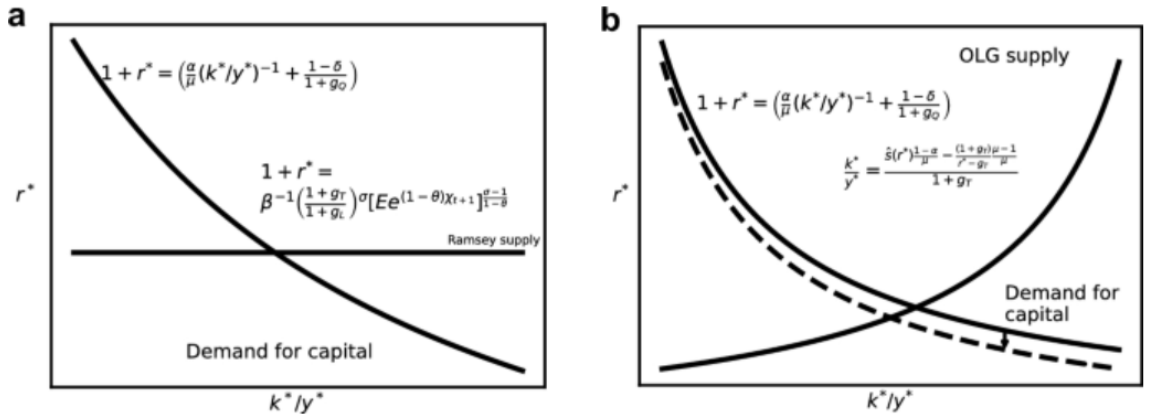
Beyond the standard framwork

- Wasn't the influnence from (relative) investment price ambigious?
  - A one-off change in the level, yes. But the above is about (changes in) the rate of *deflation. This is unambigious.*
- Macroeconomic importance for the real rate?
  - Once again, *the savings motive discussion* is at the heart of whether depreciation (physical and economic) matters.
- In a Diamond setting it matters. *Core intuition* from two period Diamond model, log utility. Only difference to discussion above would be that the first order condition for capital needs to be modified $F_K = MPK = r_t + \delta$.
  - Higher depreciation *lowers* the net marginal product of capital $\Rightarrow$lower r^*. In this setting: $\delta \uparrow \Rightarrow r^* \downarrow$ on a 1:1 basis.

Beyond the standard framework

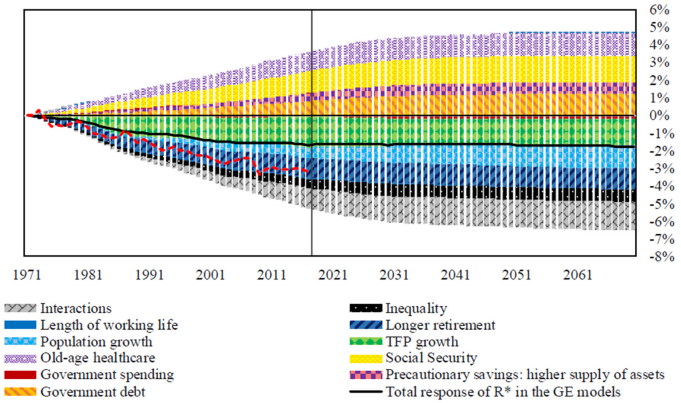
- A more realistic (calibrated) model shows, however, that higher depreciation plausibly reduced the risk-free rate by 30 basis points over the past half century (10% of the observed decline)
  - Modifier 1: Interest sensitive savings. Higher depreciation lowers savings (lower r) and reduces the supply of capital.
  - Modifier 2: Imperfect competition and portfolio effects. More depreciation increases the relative return on equity investments (dividend payments), which lowers the supply of capital.

Fig. 4



Taking stock

- Lots of suggested mechanisms. Two most often cited factors are : productivity growth and demographics. That is, a drivers behind declining real rate predicted by (essentially) all workhorse models (Solow-Swan, Diamond, RCK)
  - Trend in population growth is observed ; rising distance to the demographic transition
  - Declining productivity growth/ speed of tech progress? More controversial, but many believe technological change has slowed down (return to this later on in this course)
- Rachel and Summers (reading list) provide one attempt to gauge relative importance of a range of mechanisms, assigning first order effect to "n" and "g".



Empirical considerations

- What is the trend in observable rates? (Recall, r* is estimated.) Jorda et al tries to produce an average (portfolio and GDP weighted) return

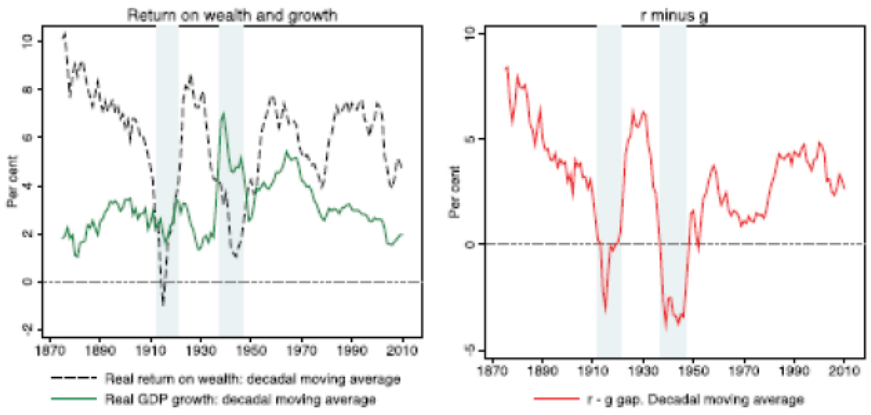
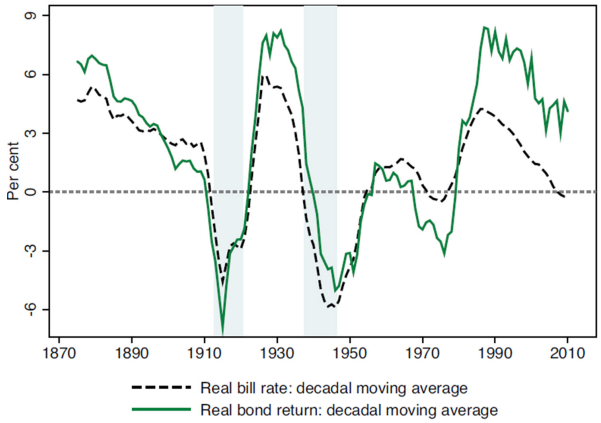


FIGURE XV
The Real Return on Wealth and Real GDP Growth

- Comprises return on housing (land), equity and government bills since late 19th century. GDP weightned. Strongly fluctuating with marked declines during wars
  - No obvious dynamic inefficiency (cf inequality lecture) judged by "r - g".

Empirical considerations

- If specializing to "safe" return (government bonds are considered "safe")



- Safe assets (bonds, bills) is what most people have in mind when they talk about declining real rates in recent decades
- Also fluctuating with peaks and valleys. As overall return: Perhaps mild tendency to decline over the period.

Empirical considerations

- Recent work by Rogoff et al (2024) provides a startling perspective. Long maturity debt obligations (sovereign debt)

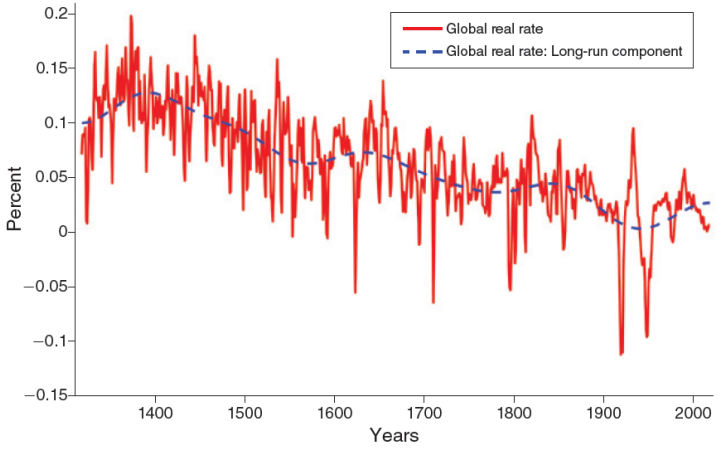


FIGURE 1. HEADLINE GLOBAL REAL RATES AND ITS LONG-RUN COMPONENT, 1318–2022

- A "super secular" decline? If true (expect serious discussion on this) it poses serious challenges. Population growth and output growth has monotonically *risen* over the period, yet the real rate has declined!
  - Key proposed explanations for recent decline seem to suffer from external validity issues. How to explain?

Learning objectives

- The natural real rate ("r*"): what it is and why it matters
  - Recieved perception: Decline since 1970 in r*.
- Ability to derive theoretical predictions regarding the long-run real rate of interest (r*) from neoclassical growth models (Solow-Swan; Diamond; Ramsey-Cass-Koopmans)
  - Differences and similarities in predictions. With and without government.
- Additional determinants discussed in Rachel and Smith (inequality, relative price of investment etc). Intuition behind the results in Dalgaard and Olsen (as in the slides)
- Key stylized facts on real rate of return (no dynamic inefficiency); the possibility of a super secular (post 15th century) decline in long maturity government bonds.

Literatur not on reading list I

- Jorda, O., S. R. Singh and A. Taylor., 2020. The Long-Run effects of Monetary Policy. Working paper.
- Laubach, T., & Williams, J. C. (2003). Measuring the natural rate of interest. Review of Economics and Statistics, 85(4), 1063-1070
- Reis, R. (2007). The analytics of monetary non-neutrality in the Sidrauski model. Economics Letters, 94(1), 129-135.